Question Bank

Subject: - Graph Theory(ECS-505)
Branch: - Computer Science
Year: - 3rd
Subject Teacher: - Ms. Payal Kansal

Unit: - 1

Year (2003-2004)

1) Prove that the sum of the degrees of the vertices of a graph is equal to twice the number of edges. Does the theorem hold for a multigraph? Justify your answer with example?

2) For the following pair of graphs, determine whether or not the graphs are isomorphic? Give the justification for your answer?

3) Proof that a simple graph with n vertices and k components can have at most (n-k) (n-k+1)/2 edges?

4) Prove that the finite connected graph is Eulerian if and only if each vertex has even degree?

5) Prove that, in a complete graph with n vertices, there are (n-1)/2 edge disjoint Hamiltonian circuits, if n is odd number >= 3?

6) Define the following with one example each :
   a) Subgraph
   b) Spanning Subgraph
   c) Homeomorphic graphs
   d) Unicursal line
   e) Arbitrarily traceable graph

Year (2004-2005)

1) Prove that in a graph the number of the vertices with odd degree is even?
2) Find a path of length 9 and a circuit of length 8 in the Peterson graph?
3) Find three Hamiltonian circuits in dodecahedron?
4) Prove that every graph with n vertices with at least n edges contains a circuit?
5) Write a brief note of 200 words or more on the travelling sales person?

Year (2005-2006)

1) Prove that the sum of the degrees of all vertices of a graph is even?
2) Prove that a simple graph with n vertices and k components can have at most (n-k) (n-k+1)/2 edges?
3) Define an Euler graph? Find an example of eulerian graph which is not Hamiltonian?
4) Define the ring sum of two graphs? Find the ring sum of the following graphs G1, G2?

5) Define the Hamiltonian path? Find an example of a non Hamiltonian graph with a Hamiltonian path?
6) Prove that a graph is an Euler graph if and only if it can be decomposed into circuits?
7) Prove that in a complete graph with n vertices there are (n-1)/2 edge disjoint Hamiltonian circuits and n >= 3?
8) Describe briefly the travelling Salesman problem?
9) Define isomorphism between two graphs? Verify whether the following graphs are isomorphic to each other?

Year (2006-2007)

1) Define a bipartite graph? Show that the complement of a bipartite graph need not to be a bipartite?
2) Discuss the Konigsberg Bridge problem?
3) Define the following with one example each :
   a) Infinite graph
   b) Hamiltonian path
   c) Component of a graph
   d) Euler graph
   e) Spanning subgraph
4) Define isomorphic graph? Draw three isomorphic graph of the following graph?

5) Differentiate, with example, a simple graph and a multigraph. Show that the maximum number of edges in a simple graph with n vertices n (n-1)/2?
6) What is the largest number of vertices in a graph with
   a) 35 edges if all vertices are of degree at least 3.
   b) 24 edges and all vertices of the same degree.

Year (2007-2008)

1) Define the degree of a vertex in a graph. Prove that the number of vertices of odd degree in a graph is always even?
2) Prove that in graph with n vertices and k components the maximum number of edges cannot exceed (n-k)(n-k+1)/2?
3) Define an Eulerian and a Hamiltonian graph? Give examples of Eulerian non Hamiltonian graph G, and Hamiltonian non-eulerian graph G2 with number of vertices >= 10?
4) Define connected graph? Prove that for a graph with exactly two vertices of odd degree, there must be a path joining these two vertices?
5) Draw a graph G with Hamiltonian path without Hamiltonian circuit with number of vertices >= 20?

Year (2009-2010)

1) Define the degree of a vertex in a graph? Prove that the sum of the degrees of all vertices of a graph in a graph is twice the number of edges in graph?
2) Define isomorphism of graphs? For the following pair of graphs , determine whether or not the graphs are isomorphic. Explain your answer?
3) Prove that a simple graph with n vertices and k components can have atmost (n-k) (n-k+1) edges?
4) Discuss travelling sales man problem?
5) Define the following with one example.
   a) Complete graph
   b) Eulerian graph
   c) Hamiltonian graph
   d) Bi-partite graph
   e) Cut points of a graph

Unit: - 2

Year (2003-2004)

1) If G is tree with n vertices then prove that it has exactly n-1 edges?
2) Explain what is meant by spanning tree? Find four spanning trees for the following graph :

3) Find the shortest path from a to z of the following graph using Dijkstra Algorithm :

4) Use the algorithm of Kruskal to find a minimum weight spanning tree in the following graph ---

5) Prove that a connected graph G is a tree if G has fewer edges than vertices?
6) Take any spanning tree in the following graph. List all the seven fundamental cut-sets with respect to this tree---

Year (2004-2005)

1) Define the pendent vertices in a binary tree? Prove that the number of the pendent vertices in a binary tree with n vertices is (n+1)/2?
2) Define a spanning tree of a graph? Find three spanning tree in the Peterson graph?
3) Write an algorithm to find the shortest spanning tree in a weighted graph?
4) Define the shortest path in a weighted graph? Describe the Dijkstra algorithm to find the shortest path m a weighted graph with vertices more than 7?

5) Define the cut set of a graph? Find five cut sets of the following graph?

Year (2005-2006)

1) Define the centre of a tree? Prove that every tree has one or two centre?
2) Prove that if in a graph G there is one and only one path between every pair of vertices is tree?
3) Define a spanning tree in a graph? Find four spanning trees In the dodecahedron graph?
4) State the two algorithms to find the shortest spanning tree in a weighted graph. Write the details of one of these algorithms?

Year (2006-2007)

1) Apply prime's algorithm to find a minimal spanning tree of the following graph?

2) Find shortest path from v1 to v8 using Dijkstra algorithm in the following graph.

3) Define spanning tree of a graph? Show that a Hamiltonian path in a graph is a spanning tree?
4) Show a tree in which its diameter is not equal to twice of the radius? Under what condition does this inequality hold? Elaborate?
5) What are the different properties when a graph G with n vertices is called a tree?

Year (2007-2008)

1) Prove that every tree has one or two centres?

2) Define a spanning tree of a graph? Find four spanning trees of the following Peterson's graph?

3) Prove that w.r.t any of its spanning trees a connected graph with n vertices and e edges has (n-1) tree branches and (e-n+1) chords?

4) Find a shortest spanning tree in a weighted graph G, using the PRIM's algorithms where G is as follows?

5) Construct a tree with 16 vertices, each corresponding to a spanning tree of a labeled completed graph with four vertices?

6) Define fundamental circuit and cut-sets. Find five fundamental circuits and fundamental cut-sets of the graph---

Year (2009-2010)

1) If G is a non-trivial tree, then prove that G contains at least two vertices of degree 1?
2) Define binary trees and discuss two important applications of it?
3) Apply Dijakstra algorithm to find out the shortest path from the vertices a to z in the following graph?

4) Use prims algorithm to find out the minimal spanning tree of the following graph?

5) Define fundamental circuits? Find the sets of fundamental circuits(four only) of the graph given above? Take any spanning tree and find it corresponding to that spanning tree?
6) Define eccentricity of the vertex and centre of a graph? Find the centre of the graph given above?

Unit: - 3

Year (2003-2004)

1) Draw a graph with
   Edge connectivity = 4
   Vertex connectivity = 3
   Degree of every vertex >= 5
2) Show that the complete bipartite graph K3,3 is non-planer?
3) In a simple connected planner graph G, there are r regions, v vertices (v>= 3) and e edges (e>1) then
   a) e >= 3*2^r
   b) e <= 3v – 6
   c) there is a vertex v of G such that degree(v) <= 5
4) Prove that a graph has a dual if and only if it is planar?
5) Show by sketching that the thickness of nine vertex complete graph is three?

Year (2004-2005)

1) Define a planar graph? Prove that for a connected planar with n vertices and e edges e <= 3n - 6 and e <= 2n – 4?
2) Write an algorithm to detect the planarity of a graph? Detect the planarity of the graph k5 and K3,3?
3) Define the dual of the graph? Show that the complete graph of four vertices is self dual? Also, if n, e and f are the number of vertices, number of edges and number of regions of a planar graph, find these numbers for the dual of this graph?

Year (2005-2006)

1) Prove that in a graph every circuit has an even number of edges in common with any cut set?
2) Define a planar graph? State and prove the Euler's theorem for a planar graph?

Year (2006-2007)

1) Define the edge connectivity and vertex connectivity of a connected graph? Find them for the following graphs—

2) Show that a complete graph kn is planar if n <= 4?
3) Draw a spanning tree of the following graph given below and list all the fundamental circuits with respect to this tree---

4) Find the dual of the following graph?

5) Prove that a graph G has a dual if and only if it is a planar?
6) Show by sketching that the thickness of eight-vertex complete graph is two?

Year (2007-2008)

1) Define the vertex connectivity and edge connectivity of a graph? Prove that for a graph G with n vertices and e edges vertex connectivity <= edge connectivity <= 2e/n?
2) Define the capacity of a cut-set? Prove that the maximum flow possible between two vertices a and b in a network is equal to the minimum of capacities of all cut-sets with respect to a and b?
3) Define a separable graph? Prove that in a non-separable graph G set of edges incident on each vertex of G is a cut-set?
4) Define a planar graph? Prove that a complete graph with five vertices is non-planar?
5) For a planar graph with n vertices and e edges prove that e <= 3n-6?
6) Define thickness and crossing number of a graph? Find thickness and crossing numbers of the graph k5 and K3, 3?

Year (2009-2010)

1) Define a planar graph? State and prove the euler's formula for planar graph?
2) Define edge and vertex connectivity of a graph? Prove that the vertex connectivity of any graph will never be more than the edge connectivity?
3) Show that the kuratowski's first (K5) and second (K3,3) are nonplanar graphs?
4) Show that a graph has a dual if and only if it is planar?
5) Define the thickness of a graph, give one example? Find the thickness of Kuratowski's first and second graph?
6) Define cut-sets? List all cut-sets with respect to the vertex pair v2, v3 in the following graph?

Unit: - 4

Year (2003-2004)

1) What is the difference between incidence and adjacency matrices? Prepare both matrices for given graph---

2) Define the term with example—
   a) Circuit matrix
   b) Cutset matrix
   c) Fundamental cut set matrix

3) Prove that m-vertex graph is a tree if its chromatic polynomial is Pm (n) = n (n-1) ^ (m-1)?
4) Define Arborescence with example? Discuss its one application? Also prove that an Arborescence is a tree in which every vertex other than root has an in-degree of exactly one?

Year (2004-2005)

1) Define a vector space associated with a graph G and its two subspaces the circuit subspace and cut set subspace? Find all the distinct bases of the circuit subspace of K5?
2) Define the circuit matrix B (G) of a connected graph G with n vertices and e edges? Prove that the rank of B (G) is e-n+1?
3) Define the adjacency matrix A (G) of a simple graph G? Prove that two graphs G1 and G2 are isomorphic if and only if A (G1) and A (G2) differ only by the permutations of rows and columns?
4) Define a k-chromatic graph? Prove that every tree with two or more vertices is 2-chromatic? Find an example of a 2-chromatic graph which is not a tree. Also, find the chromatic polynomial of a tree with n vertices?

Year (2005-2006)

1) Define a vector space for a graph G, and the circuit subspace and cut sets subspace of this vector space? Prove that the circuit subspace and cut set subspace are orthogonal to each other?
2) Define the incidence matrix, of a graph G? Prove that the rank of an incidence matrix of a connected graph with n vertices is n-1?
3) Define the circuit matrix B of a connected graph with n vertices and e edges? Prove that the rank of B is e-n+1?
4) Define the chromatic number and chromatic polynomial of a graph? Find the chromatic number and the chromatic polynomial of the following graph-----

Year (2006-2007)

1) Define basis vectors of a graph? Find the number of distinct basis possible in a cut-set subspace?
2) Define
   a) Reduced incidence matrix
   b) Fundamental circuit matrix and
   c) Fundamental cut-set matrix

   Of a connected graph? Also device the relationship between them?

3) Consider the circuit matrix (B) and incidence matrix (A) of a simple connected graph whose columns are arranged using the same order of edges. Then prove that every row of B is orthogonal to every row of A? also verify the result for the following graph-----

4) What do you mean by chromatic number and chromatic polynomial of a graph? Determine the chromatic number and chromatic polynomial of the following graphs---

Year (2007-2008)

1) Define a vector space of a graph? Find five base and number of vectors in the vector space of graph given below? Also find five cut-set vectors and five circuit vectors of this vector space?
2) Define the adjacency matrix of a graph? Find the rank of the regular graph with n vertices and with degree p (<n) of any vertex?
3) Define reduced matrix AF, fundamental circuit matrix bf and the fundamental cut-set matrix Cf of a connected graph G with n vertices and e edges. Derive the relationship among AF, bf and Cf?
4) Define the chromatic polynomial of a graph? Find the chromatic polynomial of the graph given below?
5) State and prove five colour theorem?

Year (2009-2010)

1) Define basis vectors of a graph? Show that the number of distinct basis possible in a cut-set subspace is :
   1/r! (2^r – 2^0) (2^r – 2^1) (2^r – 2^2)……. (2^r – 2^ r-1)
2) If B is a circuit matrix of a connected graph G, with e edges and n vertices, then show that the rank of B is equal to the nullity of G?
3) Prove that the rank of a cut-set matrix is equal to the rank of the graph?
4) Prove that the m-vertex graph is a tree if and only if its chromatic polynomial is

   Pm(x) = x (x-1) ^m-1.